

EE/CprE/SE 4920 STATUS REPORT 3

2/12/26 - 2/26/26

Group Number: sdmay26-40

Project Title: True Force Technologies Arcade Rack

Client: True Force Technologies

Advisor: Matt Post

Team Members/Role:

Jacob Garcia - Frontend Development

Dylan Longlett - Frontend Development, Electronics

Andy Drafahl - Team Manager, Frontend Development

Fadi Masannat - Frontend Development

Sofi Gutierrez - Frontend Development

Parnika Dasgupta - Client Liaison, Frontend Development

Colin Yuska - Electronics

Weekly Summary

This week, the team worked on and finished several mini games. Specifically, our Rocket Launch mini game is up and running. There are several other new games in the works, and the team is getting helpful gameplay feedback from users. The team also continued wiring up the arcade cabinets and ordered needed parts. The animator for the power puller game has been hired and is currently working on the animation, and music is being written for each of the games.

Past Week Accomplishments

Jacob Garcia

I have fully developed the first version of the Rocket Launch Mini game. It is currently deployed and running on the arcade cabinet and works flawlessly.

Parnika Dasgupta

I worked on 2 mini games. One of the games is done (named "What's your height"). I just need to improve the aesthetics on that. The other game is still a work in progress.

Dylan Longlett

Worked on arcade wiring and games, also look user feedback and adjusted game play mechanics.

Fadi Masannat

I worked on making my own minigame to test and experiment with what we have so far. I got more out of it than just the minigame, I got comfortable running the script and initializing an updated app instance on the Raspberry Pi. I ended up strapping the minigame as it was not as good as I wanted it to be, but I learned a lot in the process which will help me for my upcoming goals.

Sofi Gutierrez

Added more screens and "effects" to my flappy bird game, such as timers, countdown, leaderboard display, and game over.

Colin Yuska

Found the remaining electrical parts we need to finish the design from our clients expectations. In addition, ordered these parts and made plans to implement them.

Andy Drafahl

Made final revisions to the power-pull animation proposal and sent it over to the animator, got it approved with the animator and purchased the animation. Made a separate boot-up animation from scratch with sounds and music. Worked on music for the rocket game.

Individual Contributions

Name	Individual Contributions	Hours This Week	Hours Cumulative
Jacob Garcia	Finished first version of Rocket Mini game	17	22
Parnika Dasgupta	Finished one mini game (What's your height), and am working on another mini game	3	6
Dylan Longlett	Worked on arcade wiring and games, also look user feedback and adjusted game play mechanics	5	37.67
Sofi Gutierrez	Continued working on "flappy bird"	2	6
Colin Yuska	Researched and found ideal parts that we need.	5	7.75
Fadi Masannat	Drafted a minigame conceptually and on code. Decided to postpone / cancel for now.	3	10
Andy Drafahl	Revised power-pull animation doc, approved doc with animator, purchased animation, made boot up animation with sounds and music, worked on rocket game music.	6.18	8.51

Plans for the Upcoming Week

Jacob Garcia

I will continue to work on the Rocket Mini Game. I will most likely implement small quality-of-life changes and small new features.

Parnika Dasgupta

I will improve the aesthetics on "What's your height" and finish my other mini game.

Dylan Longlett

Continue to work on setting up the cash acceptor and implementing user feedback.

Fadi Masannat

We have gotten some valuable feedback from the client as he wants an extremely basic version of the game as the default entry, with all the minigames as accessible options, thus I plan on making that essential core game mode. I also would like to experiment with Jacob's game and implement some additional features and discuss those with him.

Sofi Gutierrez

I am gonna completely change the aesthetics of the game (before it was designed to mimic the same design as PowerPuller3000), add sound effects, and hopefully start testing it with the load sensors.

Colin Yuska

My plan is to wire up the cash acceptor and the receipt printer once that comes in. Also, depending on whether our Arduino hat gets shipped in time, I will add that as well. Finally, I will look into getting new boards for mounting our devices on the cabinet exterior.

Andy Drafahl

I plan to go all in on music for the various minigames, getting feedback from the team as I go. My hope is that as I get more familiar with the software, I'll get much faster at cranking out tracks.

Summary of Weekly Advisor Meeting

During this week's meeting, we demoed Jacob's mini game to our client. Our client seemed to like it very much. He suggested a few changes to the game.